

KAIHUAI DENG

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EDUCATION

Nanjing University of Information Science and Technology

2023.09 – 2027.07 (Expected)

- B.S. in Atmospheric Science, Hydrometeorology division.
- Weighted average score: 85.5 | GPA: 3.6

AREA OF INTEREST

Remote Sensing · Land–Atmosphere Coupling · Extreme Events · WRF Modeling

PUBLICATIONS

- [1] Deng, K., Huang, Z.*, (2026). **Area expansion driven intensification and evaporative amplification of global drought.** *Geophysical Research Letters*. (to be submitted as Letter)
- [2] Deng, K., Huang, Z.*, Li, H., Wu, J., (2026). **Global unequal exposure to intensifying hydrological droughts.** *Journal of Hydrology*. (Major Revision)
- [3] Wang, J.* and Deng, K. (2024). **A Model that Explains the Contrasting SST Trends in the Southern Pacific Ocean.** *STECEP*. doi.org/10.61173/xnag2k18

RESEARCH EXPERIENCE

Global 3D Hydrological Drought Tracking & Mechanism Attribution

2025.09 – Present

Independent research. Supervised by Prof. Zhongwei Huang (NUIST)

Nanjing

- Applied **EEMD** to 23-year GRACE/GRACE-FO TWS, developed a **3D drought clustering algorithm**, identifying **672** discrete global drought events with topological splitting/merging.
- Constructed $C_E = E'/(E' - P')$ evaporative contribution framework: **14.8%** of grid cells shifted from supply-deficit to compound droughts; area expansion drives **70–80%** of global severity increase ($p < 0.05$).
- Presented at **Tsinghua University DESS 2026 National Forum** (70 nationwide, English defense); Provincial Undergraduate Training Program on Innovation and Entrepreneurship (XJDC202610300604, PI).

Global Unequal Exposure to Intensifying Hydrological Droughts

2025.03 – 2025.09

Research assistant. Supervised by Prof. Zhongwei Huang (NUIST)

Nanjing

- Computed **DSI** from GRACE/GRACE-FO CSR RL06 v03 mascon TWS via EEMD; integrated GPW v4 population grids, decomposing exposure trends: climate **64.5%**, population **18.2%**, interaction **17.3%** (severe events: climate **76.1%**).
- Revealed **42.78 million people/yr** exposure increase concentrated in developing regions; most high-income regions showed stable/declining trends. Under **Major Revision** at *Journal of Hydrology*.

Fluctuation Theorem & TCWV Asymmetry

2024.09 – 2025.09

Independent research. Supervised by Prof. Jun Yin (NUIST). Discussed with Prof. Lamberto Rondoni (Politecnico di Torino)

- Extended nonequilibrium **fluctuation theorem** (Yin et al., 2024, *J. Climate*) from surface temperature to TCWV using ERA5; quantified PDF asymmetry A_{asym} across $\tau = 1\text{--}20$ yr smoothing timescales.
- Validated SST–TCWV Clausius-Clapeyron scaling ($R^2 = 0.95$, 7%/K); migrated tail-rescaling framework to precipitation, demonstrating universality. Produced manuscript draft.

WRF-based Precipitation Sensitivity over Jing-Jin-Ji2025.03 – 2025.07

Independent research. Supervised by Dr. Mengya Wang (NUIST)Nanjing

- Configured **WRF/WPS** with 3 nested domains on Linux HPC; conducted **16 sensitivity experiments** perturbing soil moisture ($\pm 10\%$ to $\pm 50\%$) and urban LULC to simulate rapid regional urbanization.
- Evaluated precipitation responses via spatial correlation, RMSE, and threat scores; identified optimal cumulus and microphysics parameterization schemes minimizing simulated biases.

Precipitation-Runoff Prediction in the Yiluo River Basin2024.06 – 2025.03

Research assistant. Supervised by Prof. Xing Yuan (LabESM, CAS)Nanjing

- Applied **EOF** to monthly precipitation interannual increments, extracting 3 PCs explaining **>90%** variance; screened key hydroclimatic predictors via correlation analysis and stepwise regression.
- Built **LSTM + interannual increment** hybrid model: TCC improved **16%**, RMSE reduced **20%** vs. statistical downscaling; CSSPv2 basin runoff forecasts achieved **NSE = 65%** for monthly streamflow.

Contrasting SST Trends in the Southern Pacific Ocean2024.03 – 2024.09

Student researcher. Supervised by Prof. George Tselioudis (Columbia University / NASA GISS)Beijing

- Analyzed **29-year multi-source satellite/reanalysis datasets** (Copernicus SST, DUACS SSH, CERES, NCEP, NSIDC SIC) to characterize SST trends across the Southern Pacific (80°W–180, 70°S–30°S).
- Derived **Ekman transport** (U_E , V_E , W_E) from wind stress to attribute the SST dipole; diagnosed a **sea ice radiative positive feedback** cycle. Published as co-first author in *STECEP* (2024).

INTERNSHIP

Climate Smart Lab, University of Prince Edward Island2026.07 – 2026.10

Research Intern. Supervised by Prof. Xander Wang (UPEI)PE, Canada

- Fully funded by **Mitacs Globalink** and **Chinese Government Scholarship (CSC)** (≤ 260 awardees nationwide). Conducting land-atmosphere coupling diagnostics and **GRACE-based groundwater trend analysis** for PEI.
- Comparing satellite TWS estimates with local well measurements to assess freshwater sustainability in a groundwater-dependent province.

Leizhou Meteorological Bureau, Guangdong2025.07 – 2025.08

Intern, Forecasting Division, Guangdong Provincial Meteorological ObservationZhanjiang

- Participated in frontline monitoring/warning for **Typhoon Danas (2025)** and **Typhoon Wipha (2025)**, tracking intensification and landfall trajectories; conducted **multi-model NWP comparison** (ECMWF, CMA, NCEP, TRAMS).
- Calibrated field hydrologic equipment (groundwater wells, flux towers, snow sensors) utilizing telemetry systems, ensuring continuous environmental data collection for disaster forecasting.

SELECTED AWARDS & HONORS

National Outstanding Undergraduate Forum. Tsinghua University, DESS (70 nationwide)Jun 2026

Chinese Government Scholarship. Chinese Scholarship Council (≤ 260 awardees nationwide)Jan 2026

Mitacs Globalink Scholarship. Mitacs, Canadian national research organizationJan 2026

Second-Class Academic Scholarship. Nanjing University of Information Science & TechonologyNov 2025

Second-Class Academic Scholarship. Nanjing University of Information Science & TechonologyNov 2024

Merit Student. Nanjing University of Information Science & TechonologyDec 2024

ADDITIONAL INFORMATION

- Programming:** Python (numpy, scipy, xarray, PyTorch), Shell, Fortran, MATLAB, Typst/LaTeX.
- Methods & Models:** EEMD, Clustering algorithm, EOF, GMM, LSTM, WRF/WPS, Xinanjiang (XAJ).
- Data & Tools:** GRACE/GRACE-FO, ERA5, GLDAS, Google Earth Engine, Linux (HPC), Git.
- Languages:** English (IELTS 7.0, C1), Chinese (Mandarin, Cantonese, Hokkien), French (A2).